

NORTHGATE 5 ESE, ND, USA

WMO: 727540

Lat: 48.9676N

Lon: 102.1702W

Elev: 561

StdP: 94.76

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 94084

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.9	-28.2	-33.6	0.2	-30.4	-30.7	0.2	-27.7	11.0	-9.7	10.1	-10.7	2.8	N/A	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	30.5	19.4	28.8	19.1	27.0	18.3	21.8	27.5	20.7	26.2	19.5	25.1	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	15.7	25.0	18.7	14.5	24.0	17.5	13.4	22.5	66.4	27.7	62.3	26.0	57.9	25.0	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.5	-35.8	34.4	2.6	2.3	-37.7	36.1	-39.3	37.5	-40.7	38.8	-42.7	40.5		
(5)			-35.2	24.3	1.9	1.5	-36.6	25.4	-37.7	26.3	-38.8	27.1	-40.2	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-12.4	-13.5	-4.6	3.4	11.0	16.2	19.0	18.1	13.1	5.1	-3.6	-11.3	(6)
(7)		DBStd	13.12	8.12	7.93	8.01	5.94	4.46	3.19	2.93	3.38	4.47	4.89	6.88	7.69	(7)
(8)		HDD10.0	3313	694	658	455	206	42	2	0	0	19	166	408	662	(8)
(9)		HDD18.3	5539	953	891	712	448	231	76	26	46	166	412	658	920	(9)
(10)		CDD10.0	925	0	0	1	8	72	189	280	251	111	13	1	0	(10)
(11)		CDD18.3	111	0	0	0	0	3	13	48	39	8	0	0	0	(11)
(12)		CDH23.3	1612	0	0	0	8	84	194	553	563	202	8	0	0	(12)
(13)		CDH26.7	449	0	0	0	1	17	33	148	191	59	1	0	0	(13)
(14)	Wind	WSAvg	3.3	4.0	3.6	3.7	3.7	3.5	3.1	2.6	2.5	2.9	3.2	3.3	3.4	(14)
(15)	Precipitation	PrecAvg	439	15	11	16	28	63	89	73	50	35	28	14	16	(15)
(16)		PrecMax	589	32	58	34	61	151	160	205	89	88	73	58	41	(16)
(17)		PrecMin	322	4	0	2	5	11	32	18	17	9	1	0	1	(17)
(18)		PrecStd	73	9	12	9	15	33	32	38	20	21	19	14	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.7	18.2	23.7	29.2	29.9	32.5	33.6	31.0	24.0	16.1	6.4	(19)
(20)			MCWB	2.3	2.3	10.2	12.5	15.8	19.6	22.1	19.3	18.4	14.4	9.5	2.5	(20)
(21)		2%	DB	3.0	2.5	13.2	19.7	25.5	27.3	30.0	30.9	28.3	19.9	11.4	3.2	(21)
(22)			MCWB	1.0	0.8	7.3	10.1	14.3	18.2	21.2	19.0	16.8	12.1	6.6	0.9	(22)
(23)		5%	DB	1.5	0.2	8.9	17.0	23.3	25.6	28.2	28.7	25.8	16.7	8.3	1.3	(23)
(24)			MCWB	0.0	-0.9	5.5	8.5	13.1	17.2	20.6	18.9	15.9	9.9	4.7	-0.4	(24)
(25)		10%	DB	-0.6	-1.8	5.1	13.9	20.7	23.9	26.5	26.5	22.1	13.4	5.4	-1.0	(25)
(26)			MCWB	-1.7	-2.7	2.6	7.0	11.9	16.4	19.1	18.1	14.4	8.6	2.9	-2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.9	10.7	12.9	18.0	22.3	24.8	22.2	20.8	15.3	9.6	3.3	(27)
(28)			MCDWB	5.2	5.7	15.7	22.3	24.5	26.1	28.6	28.5	27.6	23.3	15.7	6.6	(28)
(29)		2%	WB	1.3	1.3	7.7	10.2	16.0	19.9	22.5	21.1	18.0	13.2	6.8	1.4	(29)
(30)			MCDWB	2.9	3.0	12.3	17.9	23.3	25.1	27.6	27.5	25.7	17.9	10.6	3.6	(30)
(31)		5%	WB	0.3	-0.1	5.7	8.6	14.4	18.4	21.2	20.0	16.5	10.8	4.5	0.1	(31)
(32)			MCDWB	1.7	1.0	9.7	15.5	20.8	23.7	26.5	26.6	22.5	16.1	7.2	1.9	(32)
(33)		10%	WB	-1.2	-2.0	3.0	7.1	13.1	17.2	20.1	18.8	15.1	9.0	2.7	-1.5	(33)
(34)			MCDWB	0.0	-1.1	5.5	12.4	18.5	22.2	25.1	25.3	20.2	13.4	4.6	-0.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	10.1	10.6	12.8	13.5	12.1	13.2	14.6	13.9	11.9	10.3	9.2	(35)
(36)			MCDBR	10.6	9.9	15.6	18.9	17.9	15.5	15.4	17.9	19.7	17.9	14.8	11.3	(36)
(37)			MCWBR	9.3	7.8	10.1	10.3	8.1	7.9	7.9	8.0	9.6	10.1	9.5	8.5	(37)
(38)		5% WB	MCDBR	11.1	8.8	15.4	17.6	14.4	13.0	13.9	15.4	15.8	16.8	13.5	10.4	(38)
(39)			MCWBR	9.7	7.3	10.3	10.0	7.2	7.4	8.0	7.5	8.3	9.9	9.2	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.255	0.280	0.328	0.342	0.359	0.368	0.379	0.338	0.292	0.257	0.236	(40)	
(41)		taud	2.356	2.397	2.473	2.399	2.410	2.400	2.397	2.345	2.432	2.535	2.501	2.437	(41)	
(42)		Ebn at Noon	822	917	946	922	918	900	886	858	865	861	826	812	(42)	
(43)		Edh at Noon	66	80	89	108	111	114	112	113	93	71	57	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.51	2.61	3.88	4.94	5.81	6.19	6.65	5.38	4.00	2.45	1.45	1.08	(44)	
(45)		RadStd	0.14	0.25	0.33	0.33	0.41	0.43	0.31	0.38	0.34	0.27	0.20	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page